

Prakhar Batwal

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Education

Parul University

Bachelor of Technology in Computer Science & Engineering

Vadodara, India

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Technical Skills

- **Generative AI & Agents:** LangGraph, Agentic Workflows, RAG, LangChain, Gemini LLMs, OpenAI API, Vertex AI, Prompt Engineering
- **Machine Learning & DL:** TensorFlow, Keras, Scikit-Learn, FAISS, SentenceTransformers, XGBoost, LightGBM, SHAP, NLTK, TF-IDF
- **Frameworks & MLOps:** FastAPI, Django, Spring Boot, MLflow, DagsHub, Docker, GitHub Actions, Git, Google Colab
- **Languages & Databases:** Python, Java, JavaScript (ES6+), SQL (MySQL, PostgreSQL), MongoDB Atlas, Redis, SQLite
- **Core CS Fundamentals:** Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), DBMS, Operating Systems

Projects

Sarthi – AI Multi-Agent Software Engineering Platform

GitHub | Live

Python · FastAPI · LangGraph · Redis · MongoDB Atlas · Next.js · Docker · Gemini LLMs · Google Cloud Run

- Architected a cloud-native **distributed AI system** using **28 specialized LangGraph agents** across **7 pipeline stages** (including requirement analysis, code synthesis, validation, and testing) to produce full-stack applications from natural-language inputs.
- Engineered **memory-aware multi-agent collaboration** with **Redis-backed state persistence**, reducing inter-agent latency by **40%** across **100+ concurrent** pipelines; integrated **Gemini LLMs** and **MongoDB Atlas** for structural artifact versioning.
- Containerized microservices via **Docker** and deployed to **Google Cloud Run** for stateless horizontal scalability, leveraging an async **FastAPI** backend to reduce execution bootstrapping time down to under **10 minutes**.

Goodreads Recommendation System – Neural Search & Two-Tower Engine

GitHub

Python · TensorFlow/Keras · FAISS · SentenceTransformers · FastAPI · Docker · GitHub Actions

- Achieved **Hit Rate@10 of 25.2%** and **MRR 0.1078** by training a two-tower collaborative filtering network on 200K+ user ratings, projecting users and catalog items into a shared 64-D latent vector space.
- Replaced $O(N)$ brute-force item retrieval with a optimized **FAISS IndexFlatIP** index backed by **SentenceTransformers** real-time encoding, cutting algorithmic retrieval complexity to $O(\log N)$ approximate nearest neighbor (ANN) lookups.
- Refactored the data vectorization pipeline to embed unique catalog records once rather than per-transaction rows, dropping **CPU preprocessing time by 95%** and completely eliminating memory pressure at scale.

TruthLens – AI News Credibility & Bias Analyzer

GitHub

Python · Scikit-Learn · TF-IDF · NLTK · OpenAI API · FastAPI · JavaScript · pytest

- Trained a **dual-output classifier** (binary credibility assessment + 5-class political bias classification) utilizing **TF-IDF** feature extraction and grid-search sweeps in Scikit-Learn, optimizing F1-scores across highly imbalanced news corpora.
- Layered **NLTK VADER** for sentence-level sentiment scoring and integrated async **OpenAI API** prompt pipelines for token-level linguistic bias breakdowns mapped directly alongside core classifier confidence scores.
- Surfaced real-time text predictions through a responsive **JavaScript/HTML5** analytical dashboard displaying bias distribution charts, confidence heatmaps, and per-token sentiment highlights for non-technical users.

Network Security – Phishing Detection End-to-End ML Pipeline

GitHub

Python · Scikit-Learn · FastAPI · MongoDB · MLflow · DagsHub · PyYAML · GitHub Actions

- Architected an **end-to-end modular pipeline** across 6 continuous stages (Ingestion → Validation → Transformation → Training) handling **88,000+ network records**; ensemble classifiers hit a **97.3% F1-score**.
- Integrated **MLflow + DagsHub** for versioned tracking across Random Forest, Gradient Boosting, and AdaBoost hyperparameter sweeps, enabling full optimization reproducibility and 1-command re-training workflows from 30+ checkpoints.
- Designed a robust **PyYAML schema validation** layer to capture data drift and feature anomalies pre-training, applying **KNN Imputation**, robust scaling, and automated testing via **GitHub Actions** CI/CD.

Achievements & Certifications

- **Elevate Hackathon:** Ranked **Top 10 out of 700+ teams** (Top 1.4%); developed a production-ready system in 48 hours.
- **Competitive Programming:** Solved **400+ problems** on **LeetCode** and **CodeChef** across graphs, trees, DP, and concurrency.
- **Certifications:** Completed **MongoDB Certified Developer** Path and **AWS Cloud Practitioner Essentials**.
- **Leadership & Impact:** Led a 4-person team across 2 full-lifecycle projects; manages a ML/Full-Stack **YouTube Tech Channel**.